Technische Universität Braunschweig

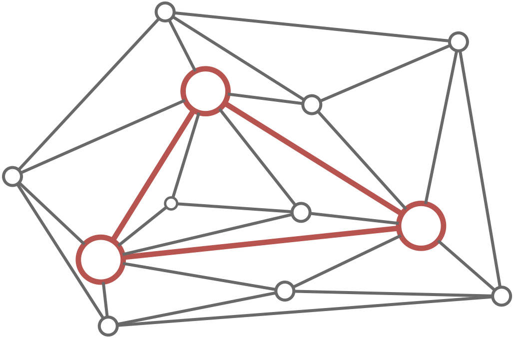
Institute of Operating Systems and Computer Networks
Algorithms

Algorithmische Methoden zu gradbeschränkten Triangulierungen

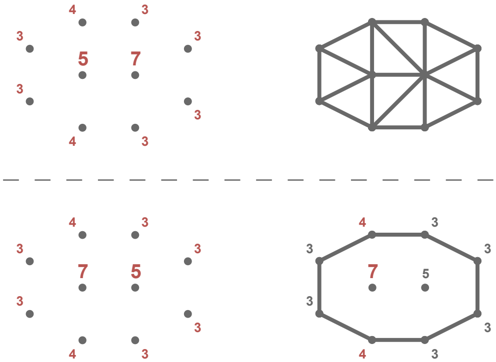
Fabian Alich, 3. September 2025

Praktische Relevanz

FA: keine Ahnung was ich sonnst noch Praktisch sagen soll.



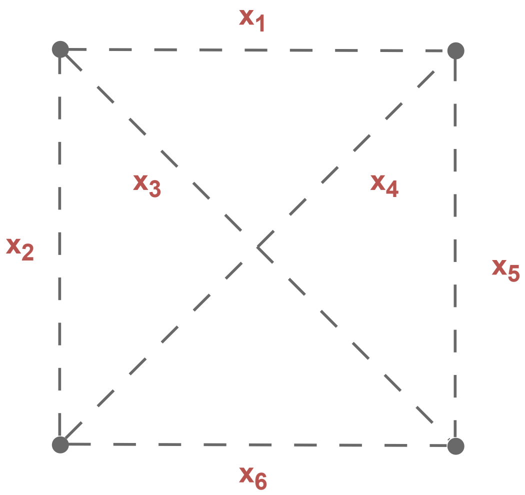
Das Problem



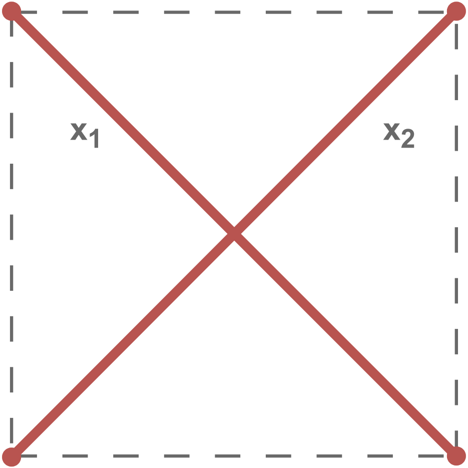
Teil I

Lösungsansätze

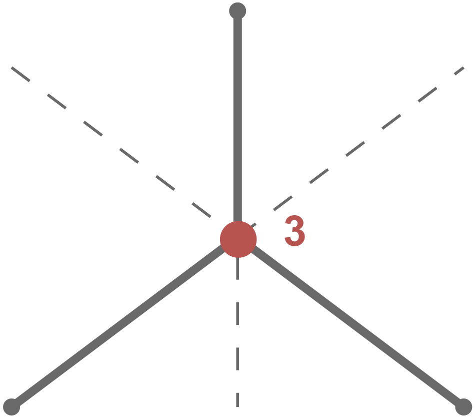
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung



Maximale Kanten

FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \overset{!}{=} \sum \text{Sollgrad}$$

Gradbeschränkung

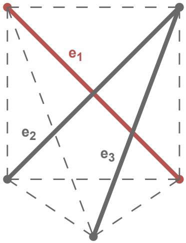
FA: Durchschnitt

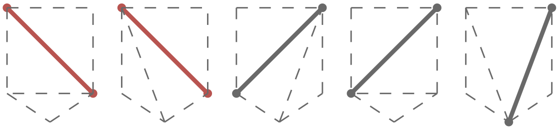
Gradbeschränkung

FA: Max min

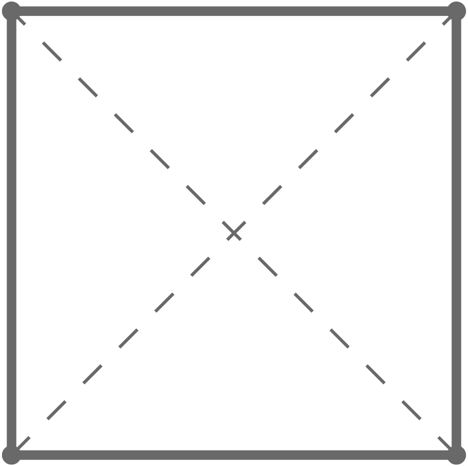
alternative Kantenbedingung

FA: sollte ich hier eine Refernz auf euer Paper machen?

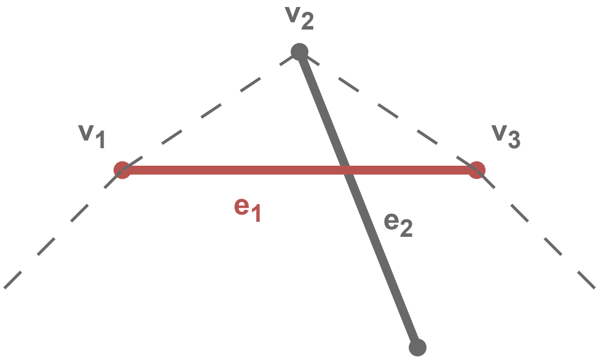




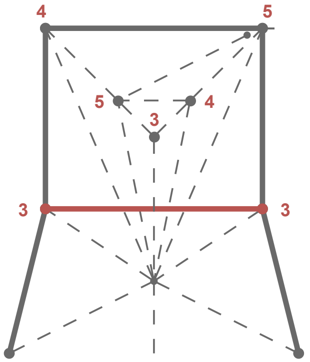
Hülle Fixieren

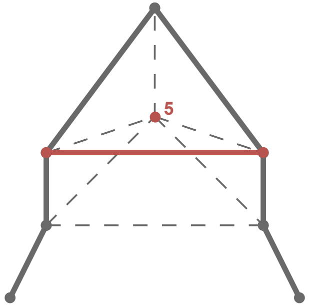


Kanten Fixieren / Ausschließen

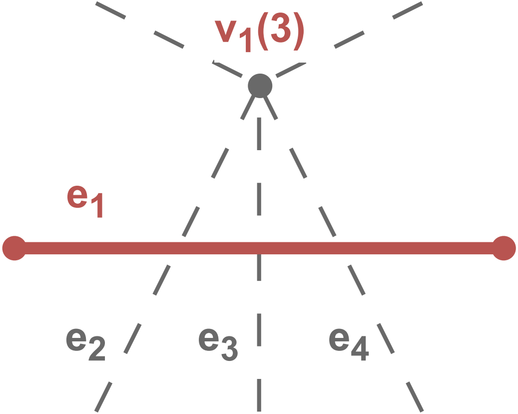


Kanten Ausschließen

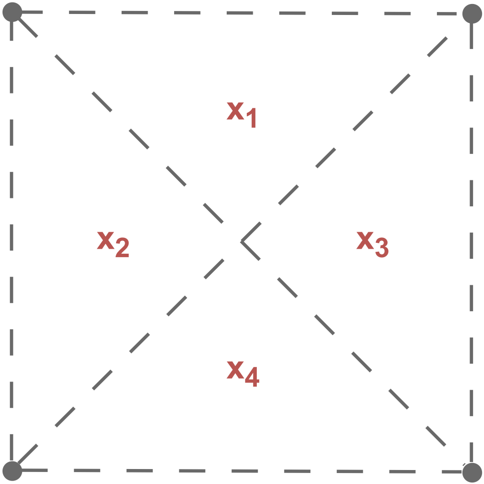




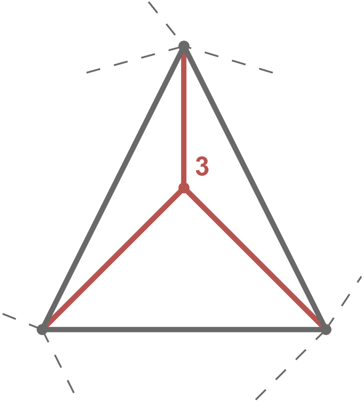
Kanten Ausschließen

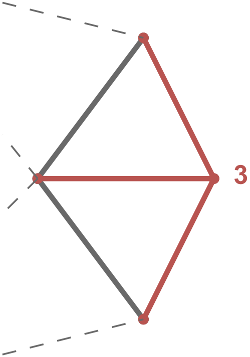


Dreiecksansatz

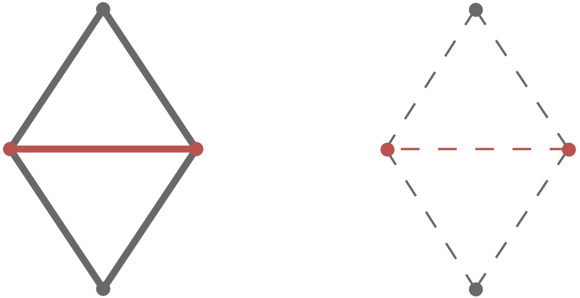


Gradbeschränkung

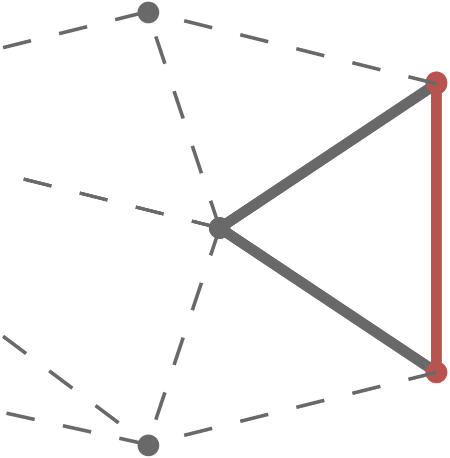




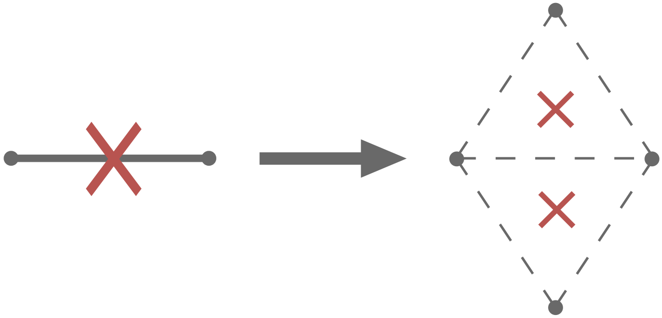
Überschneidungsbedingung



Überschneidungsbedingung



Dreiecks Ausschließen



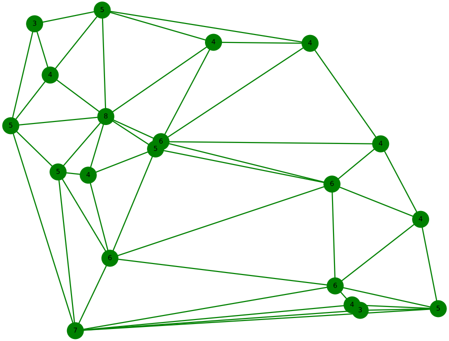
Subset

FA: ich habe jetzt subset in den Diagrammen drin, ist aber schlecht. Soll ich das dann hier erklären oder wie mache ich das am besten? Einfach nur auf die Arbeit verweisen?

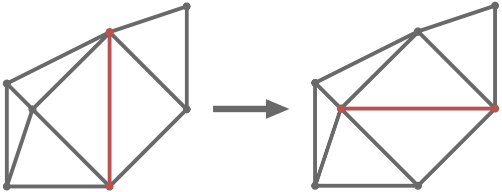
Teil II

Instanzgenerierung

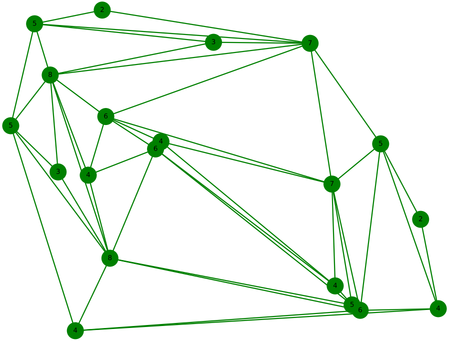
Delaunay



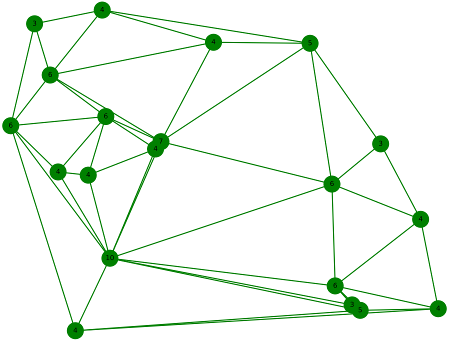
Flip



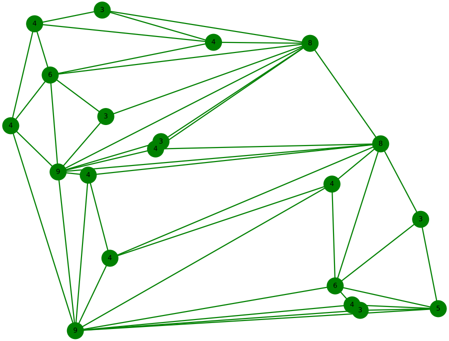
Delaunay-Flips



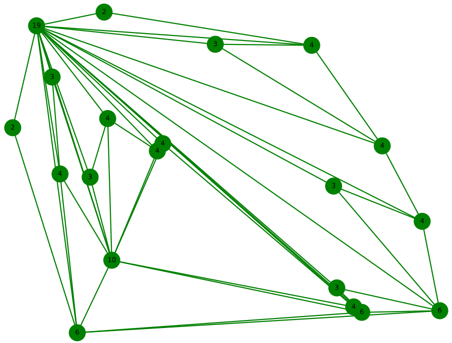
Greedy



Random



Iterativ



Move

FA: Bild erstellen

Change

FA: Bild erstellen

Teil III

Auswertung

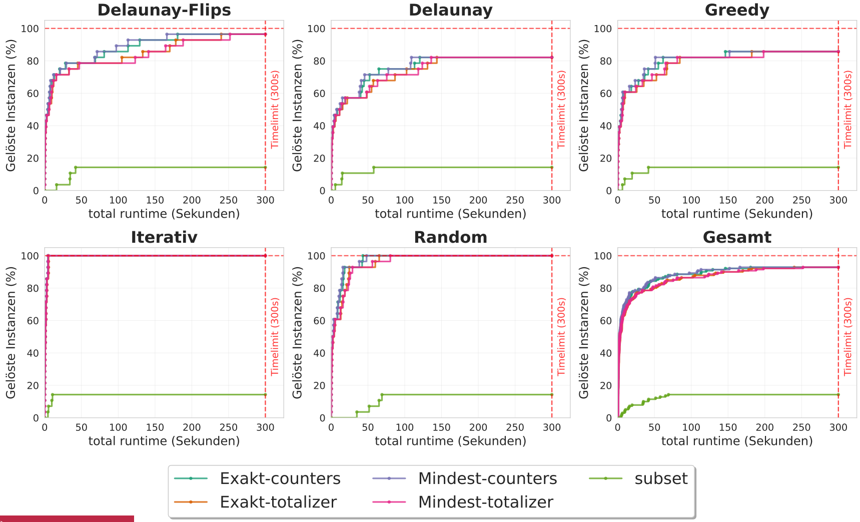
Solver

Gurobi

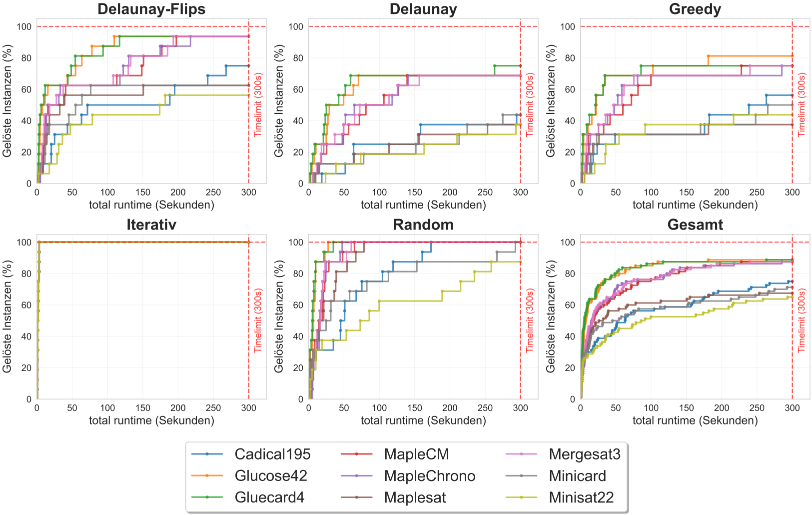
OR-Tools

PySAT

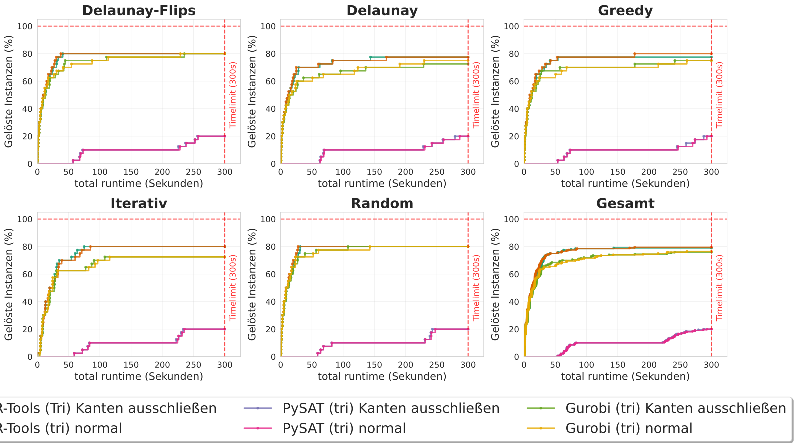
Grad Codierung



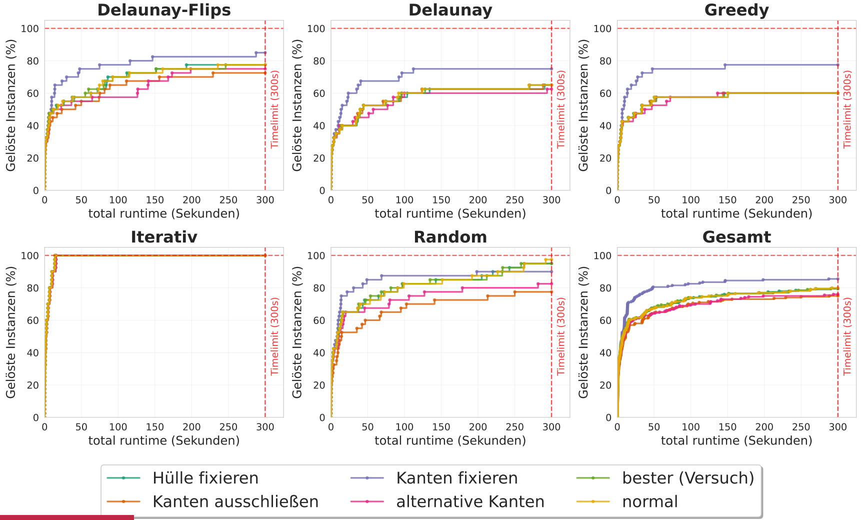
PySAT Solver



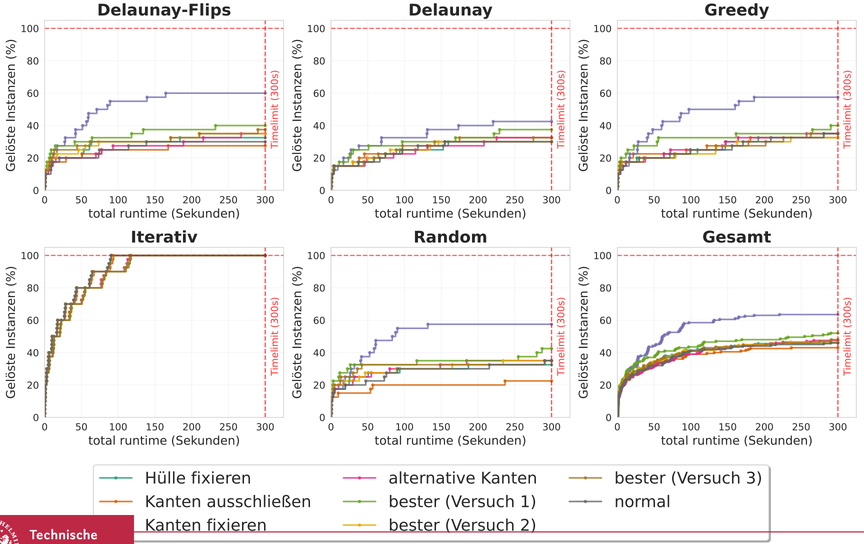
Dreiecke Bedingungen



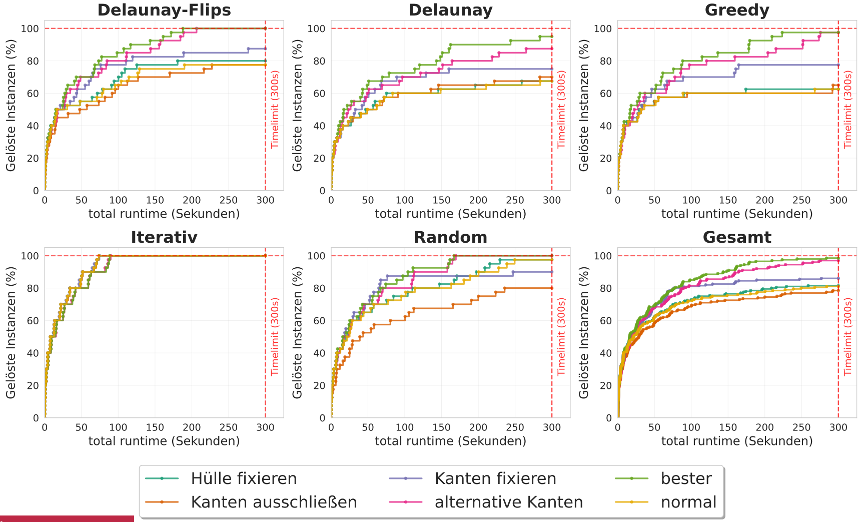
PySAT Bedingungen



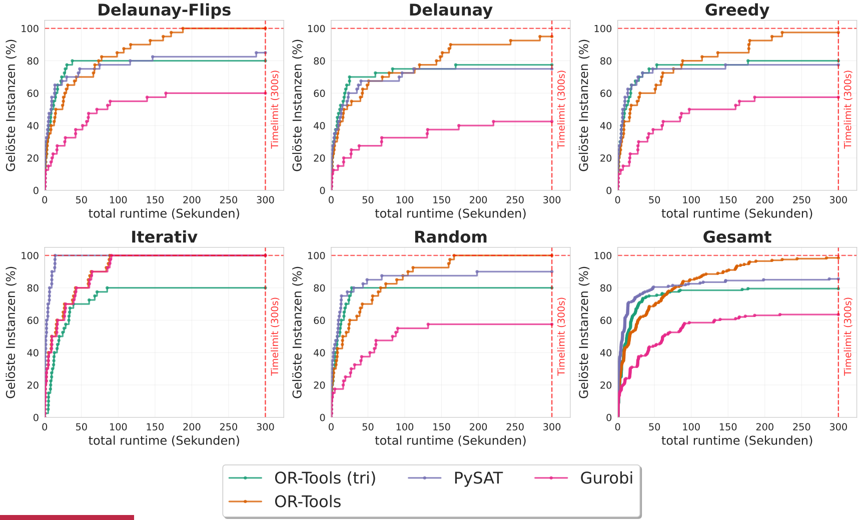
Gurobi Bedingungen



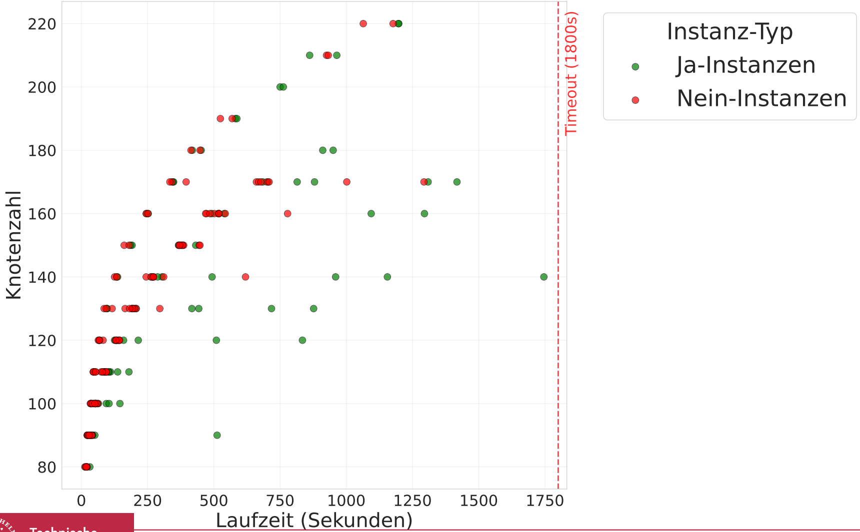
OR-Tools Bedingungen



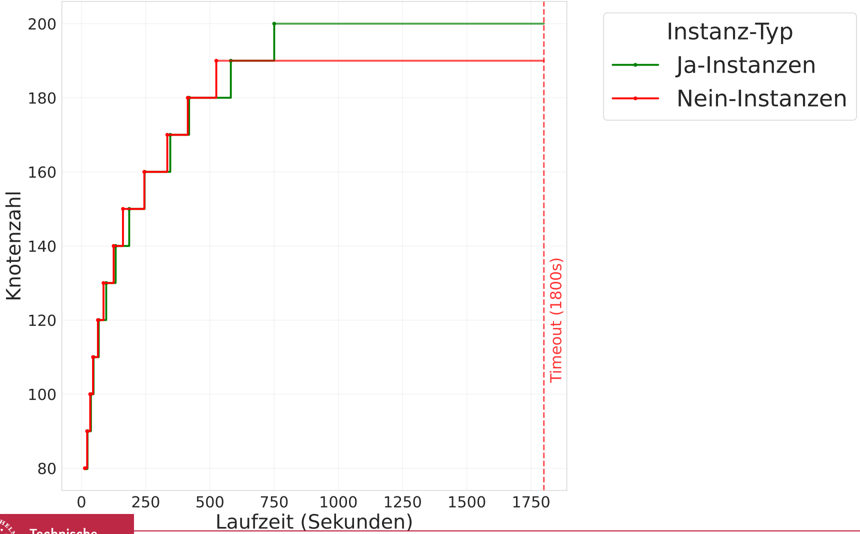
Solver Vergleich



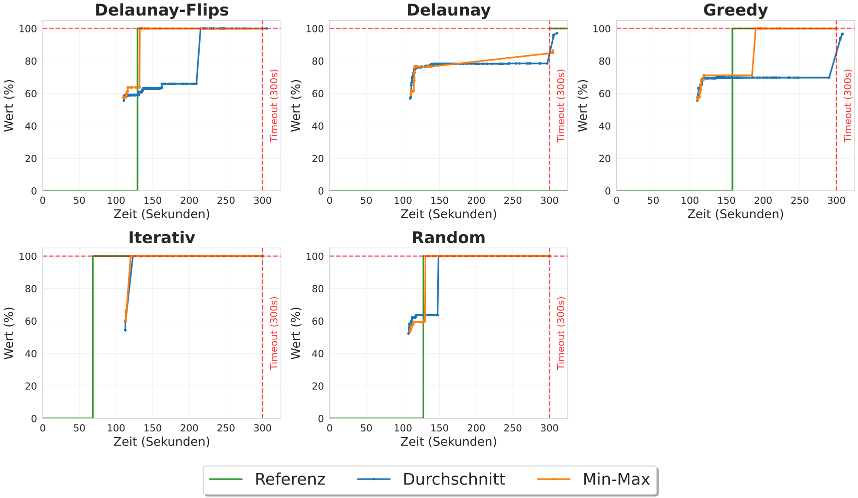
Größerer Timeout



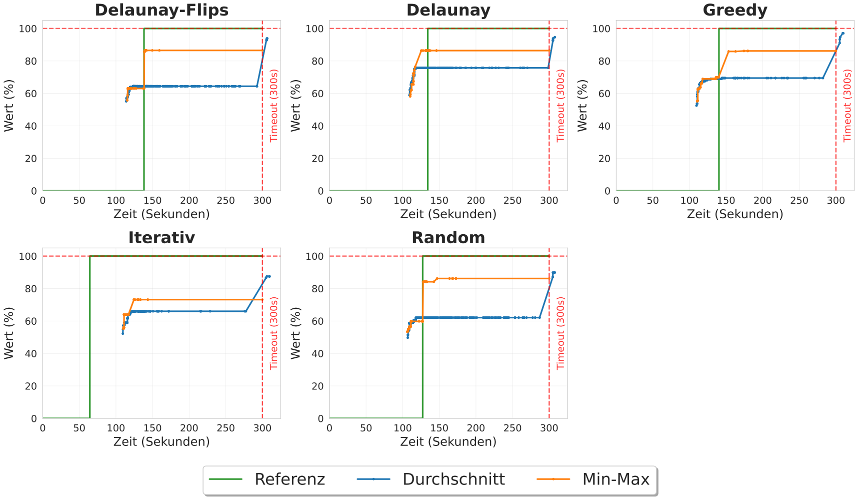
Ja Nein getrennt



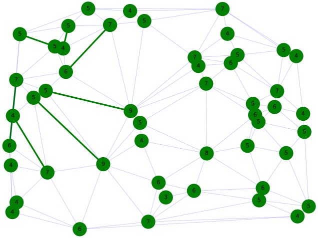
Ja Instanz

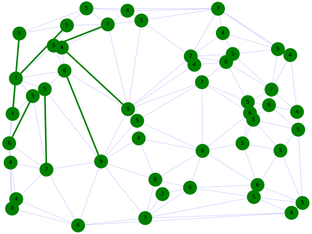

Delaunay-Flips
Delaunay
Greedy
Iterativ
Random
Wert (%)
Zeit (Sekunden)
Timeout (300s)
0
20
40
60
80
100
0
50
100
150
200
250
300
Referenz
Durchschnitt
Min-Max

Nein Instanz

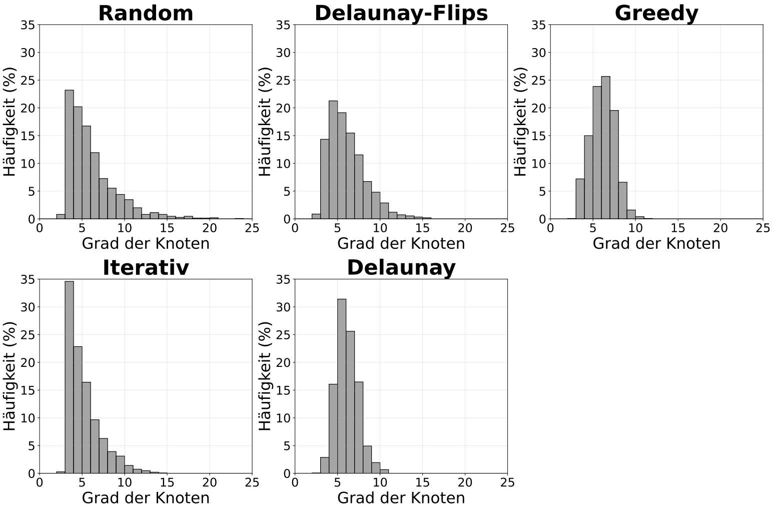


Anzahl Lösungen

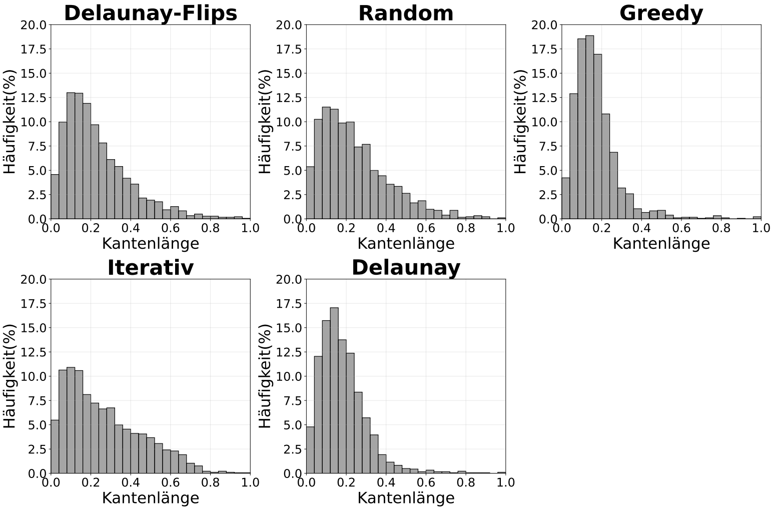




Gradverteilung



Kantenlängen



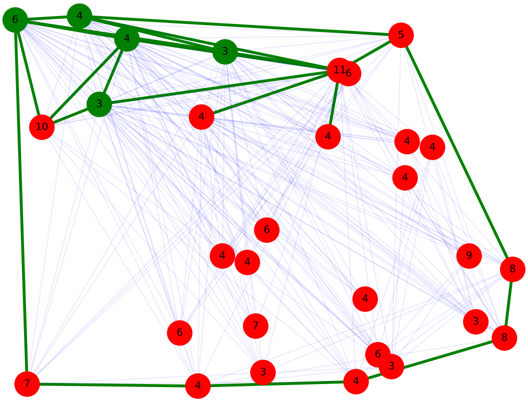
CaDiCaL

CaDiCaL Simplified Satisfiability Solver

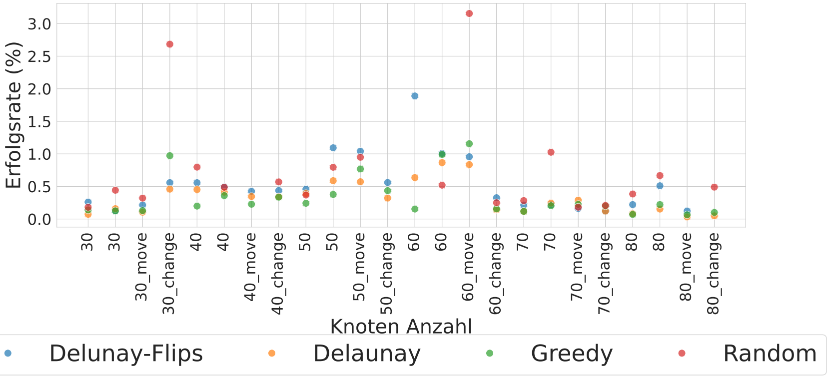
Teilinstanz

FA: Isolierte Kante Teilinstanz einfügen

Teilinstanz



Propagation



Teil IV

Fazit und Ausblick

Fazit

OR-Tools

Hülle fixieren

Kanten Fixieren

alternative Kantenbedingung

Fazit

FA: Bild von Instanze mit 220 Knoten

Fazit

FA: Bild von Optimierungsproblem mit 97 Prozent

Fazit

FA: Bild wo eine Propergation möglich war

Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierikeit Instanzen
- Optimierungen Kombinationen
- Propergation erweitern
- normalverteile Triangulierung
- Problem abwandeln
  - Kanten vorgeben
  - Polygone